ALICE L WALTON SCHOOL OF MEDICINE

Cardiac Cycle

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Adapted from: Cardiac Cycle

Listen to this Brick
0:00 / 29:32 1x

Learning Objectives (3) Versions

After completing this brick, you will be able to:

1. Explain systole, diastole, and relate to the function of the atrioventricular valves, semilunar valves, and great vessels.
2. Explain how the pressure and volume in the heart chambers and major vessels change in each of the seven phases of the cardiac cycle.
3. Correlate th al exam findings: pulse, first and second heart sounds, and the wave forms of the atrial pressure.

CASE CONNECTION

Mr. J, a 68-year-old man, presents to the clinic with exertional shortness of breath, chest discomfort, and occasional dizziness. On examination, you note a harsh systolic murmur best heard at the right upper sternal border, which radiates to the carotids. His blood pressure is 130/85 mmHg, and his pulse is slow and delayed (pulsus parvus et tardus). He is diagnosed with aortic stenosis, where it is more difficult for the ventricle to pump blood through this stiffened aortic valve.

As you read through the brick, reflect on the following questions:

Which phase of the cardiac cycle is prolonged in aortic stenosis? How does increased afterload affect stroke volume and cardiac output? What physical exam findings correlate with specific cardiac cycle phases?

GO TO CONCLUSION

What Is the Cardiac Cycle?

We all know that the heart pumps blood. But what does this really mean? How does the heart manage to take in gently flowing blood and then expel it at tremendous force? The answer lies in the cardiac cycle, a series of carefully timed events that move blood through the four chambers of the heart. These are the events that oc

We'll begin our discussion with a big-picture view of what happens to the heart chambers and valves during the cardiac cycle (also called the Wiggers diagram). Then we will define the seven phases of the cardiac cycle and discuss how the pressure and volume of the blood in the heart change during each phase. We will conclude with key exam findings and how each phase is represented on an electrocardiogram (ECG).

Systole and Diastole (LO1)

During a single normal cardiac cycle, the ventricles contract (this is called **ventricular systole** or simply systole) and relax (**ventricular diastole** or simply diastole). The atria also contract and relax, but before the ventricles do. The sequence of contracting and relaxing occurs like this:

- Atria contract (blood is pushed from atria into ventricles)
- Ventricles contract (blood is pushed from the ventricles into the aorta and pulmonary artery)
- Ventricles and atria both relax (blood flows through the atria and fills the ventricles)

What is the sequence of contraction during a normal cardiac cycle?

Valves and Vessels

All of this activity would be fruitless without one-way valves to keep blood flowing in the right direction (Figure 1).

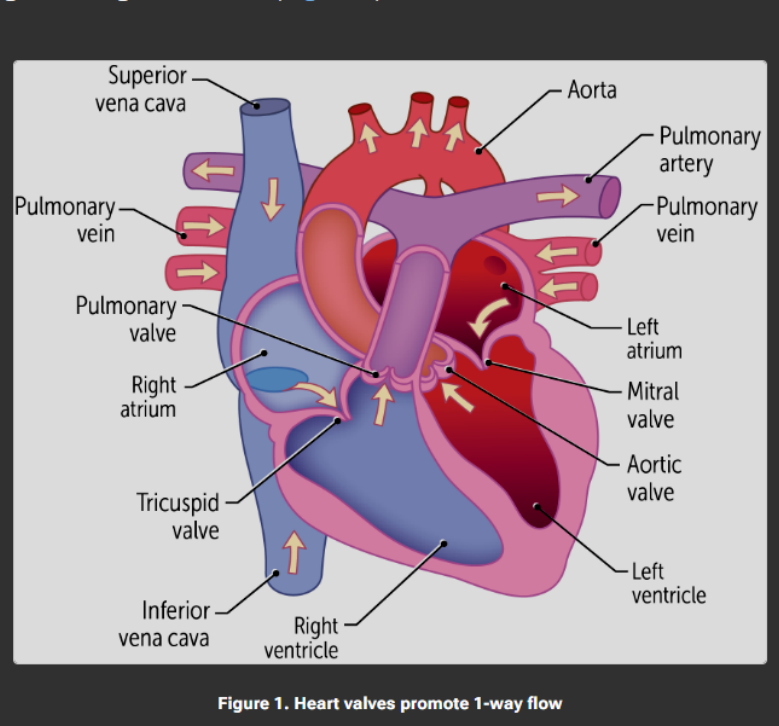


Figure 1. Heart valves promote 1-way flow

The **atrioventricular valves** (tricuspid and mitral valves, found between the atria and ventricles) remain open during the filling of the ventricles—and during the contraction of the atria—but they close when the ventricles contract so that blood doesn't flow back into the atria.

Likewise, the **semilunar valves** between the ventricles and their **great outlet vessels** (the pulmonary artery and the aorta) are open during ventricular contraction. But they close during ventricular diastole when filling occurs, so that blood doesn't flow backwards from these great vessels.

What Are the Phases of the Cardiac Cycle? (LO2)

As you might have guessed, there is more to the **cardiac cycle** than just systole (contraction) and diastole (relaxation). To really understand what's going on in the cardiac cycle, you need to understand its **seven phases**: atrial systole, isovolumetric contraction, rapid ejection, reduced ejection, isovolumetric relaxation, rapid ventricular filling, and reduced ventricular filling (Figure 2). Note: a similar figure can be drawn for the right side of the heart, albeit with lower pressures. Sometimes we hear the cardiac cycle referred to as five phases; in those discussions, people are combining the two ejection and the two filling phases, which are related but separate phases.

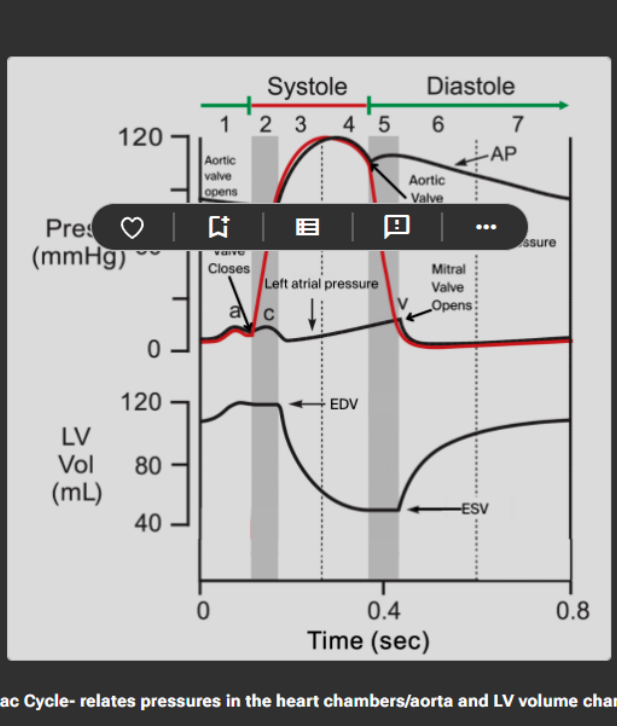


Figure 2 Cardiac Cycle- relates pressures in the heart chambers/aorta and LV volume changes over time

Remember that this is a cycle, so the phases run continually in sequence unless something happens to interrupt it. Let's look at each of the seven phases more closely.

1. Atrial Systole

This phase occurs towards the end of **ventricular diastole** [dashed green line] in Figure 2) and marks the beginning of a new cycle. During this phase, the **atria contract** (note the pressure increase marked "a" on the left atrial pressure line in Figure 2), pushing most of the blood they contain into the ventricles. If you think about how much smaller the atria are than the ventricles, then it seems like atrial contraction wouldn't provide enough blood to really fill up the ventricles. And that's true! Atrial contraction, or atrial kick, during this phase "to ventricles.

Where does the blood already present in the ventricles come from? At the end of a normal cardiac cycle, the ventricles are relaxed and passively filling with blood, since the mitral and tricuspid valves between the atria and the ventricles are open and the atrial pressure slightly exceeds the ventricular pressure. In fact, at the end of a normal cardiac cycle shown here, the ventricles are already mostly full. When a new cycle starts, the atrial contraction phase just pushes the little blood that remains in the atria into the ventricles.

During atrial systole, atrial contraction causes the ventricles to increase in pressure (slightly) and volume. At the end of this phase, the atria have emptied and the ventricles are at their highest volume. We call the volume in the ventricles just prior to their contraction the **end diastolic volume** (EDV). Here, an increased EDV (also increases preload) will stretch the ventricle and cause a stronger force of contraction.

When the heart beats at a normal (resting) rate, atrial contraction provides only about 10%-15% of the blood that fills the ventricles. Up to 90% of ventricular filling results from passive venous return during diastole, not from the contraction of the atria.

When the heart beats more quickly, both systole and diastole shorten, but diastole shortens more than systole, so the heart has less time to fill with blood. Because of this, in **tachycardia**, atrial contraction becomes more important for filling the ventricles. At maximal heart rates, atrial contraction provides up to 40% of the blood that fills the ventricles.

In which phase of the cardiac cycle do the ventricles get to the end diastolic volume (EDV)?

2. Isovolumetric Ventricular Contraction

This phase marks the beginning of **ventricular systole**. It's the period of time during which the ventricles contract but no blood moves out into the great vessels (aorta and pulmonary artery) because the pressure in the great vessels is still higher than in the ventricles, and the aortic and pulmonary valves are still closed. This is why the phase is called isovolumetric. The sole purpose of this phase is to increase pressure in the ventricles so ventricular ejection can occur.

Important valve movements bookend the isovolumetric contraction phase of the cycle:

- After the atria contract to end ventricular filling, the ventricles contract.
- As the ventricles contract, the pressure in the ventricles becomes greater than the pressure in the atria, so the **atrioventricular valves** close.
- At the end of isovolumetric contraction, the pressure in the ventricles builds to the point where it exceeds the pressure in the great vessels, and the **semilunar v** umetric contraction phase itself, these valves are closed.

How would atrial fibrillation, irregular and rapid contraction of the atria, affect ventricular filling?

3. Rapid Ventricular Ejection

In this phase, the ventricles begin to eject blood into the great vessels. How does this happen? When the pressure in the ventricles exceeds the pressure in these arteries, the semilunar valves open and blood is rapidly ejected. The volume of the ventricles rapidly decreases during this phase and the pressure in the great vessels rapidly increases as blood is ejected into them.

During this phase, the atrioventricular valves remain closed because of the high pressure in the ventricles.

4. Reduced Ventricular Ejection

This phase is similar to rapid ejection in that blood is still being ejected from the ventricles. However, as the ventricles begin to relax, the ventricular volume decreases and the force of ejection diminishes—which is why it's called reduced ejection. You may be wondering, what's the difference between this phase and rapid ejection? The difference is that blood is ejected more slowly because the pressure gradient from the ventricles to the arteries is reduced. Blood still continues to leave the ventricles due to the kinetic energy or inertia of the blood (for a short time) even when the pressure in large arteries exceeds the ventricular pressure.

The end of this phase marks the end of ventricular systole; here, the volume in the ventricles is at its lowest, and it is termed the **end-systolic volume** (ESV). Remember the end diastolic volume (EDV) we talked about in the atrial systole phase? That is the point at which the volume in the ventricles is at its highest. These are both important volumes, because they help us determine the **stroke volume** (SV), or the amount of blood that is ejected from the ventricles during a single cardiac cycle. The stroke volume is the difference between the EDV and the ESV (SV = EDV – ESV). Stroke volume is important because it is a key component of the cardiac output, the central measure of cardiac performance.

In which phase of the cardiac cycle do the ventricles get to the end-systolic volume (ESV)?

How does stroke volume relate to end-diastolic volume (EDV) and end-systolic volume (ESV)?

5. Isovolumetric Ventricular Relaxation

This phase encompasses the period during which the ventricles are relaxing and the volume in the ventricles does not change. The sole purpose of this phase is to decrease pressure in the ventricles so ventricular filling can occur later. As the ventricles relax, the pressure within their chambers decreases. The pressure in the great vessels is now greater than ventricular pressure; this closes the aortic and pulmonary valves.

The pressure in the v this period, but the volume remains constant because the atrioventricular valves are still closed.

6. Rapid Ventricular Filling

At the end of the isovolumetric ventricular relaxation phase, the ventricles reach a point at which their internal pressures fall below the pressures of their corresponding atria. When that happens, the atrioventricular valves open and blood rapidly enters the ventricles.

Remember from our discussion of atrial contraction that the ventricles fill primarily from passive atrial pressure. The venae cavae and the pulmonary veins have enough pressure to fill the ventricles without atrial contraction because of the large buildup of volume resulting from continued venous return. The atria store returning blood while the ventricles are contracting, allowing blood to flow continuously in the system. This phase is responsible for 85% to 90% of ventricular filling (passive), while atrial systole actively contributes the remaining 10% to 15% of blood to the ventricles.

7. Reduced Ventricular Filling

Just as we divided systolic ejection into rapid and reduced phases, we also divide ventricular filling into rapid and reduced phases. When the ventricles fill with blood, the pressures in the ventricles increase to the point that they are almost the same as the pressures in the atria. When this happens, the filling slows dramatically. The only significant filling that can occur beyond this point is when the atria contract in the next phase, which brings us back to the start of the cardiac cycle.

What Are Key E hases of the Cardiac Cycle? (LO3)

When you see a patient, it's helpful (and sometimes essential) to get your bearings and understand what part of the cardiac cycle you are examining. The best ways to do this are by using the **pulse, the heart sounds, and the jugular venous pressure.** We will review these briefly here.

Pulse

The pulse can be palpated at a number of places—the radial artery (wrist), femoral artery (groin), and carotid artery (neck) are the easiest. Figure 3 shows an aortic pressure tracing, which is a waveform of the pressure in the aorta during the cardiac cycle. Note that the palpated pulse (highest point on the curve) will correspond to the rapid and reduced ejection phases of systole.

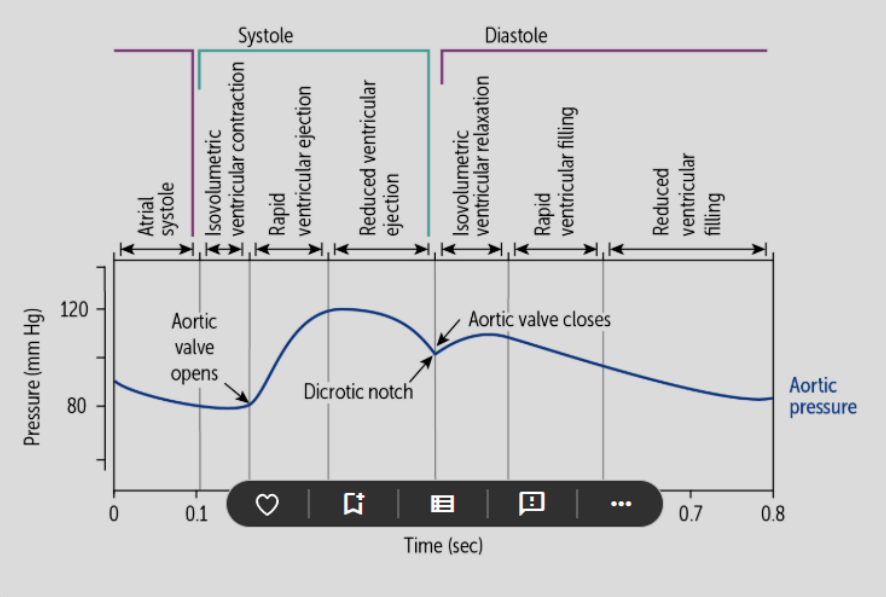


Figure 3. Aortic pressures contribute to pulse

Also note the dicrotic notch, an upward rebound of the pulse tracing that corresponds to a rebound of pressures in the elastic aorta just after closing of the aortic valve. You can sometimes see it in the carotid artery, and it corresponds to isovolumetric relaxation.

Heart Sounds

The **first heart sound** (S1) corresponds to closure of the mitral and tricuspid valves, which marks the end of atrial systole and the start of isovolumetric contraction. The **second heart sound** (S2) corresponds to the closure of the aortic and pulmonary valves, which marks the end of the reduced ventricular ejection phase and the start of isovolumetric relaxation. Heart sounds are covered in a separate brick.

Jugular Venous Pressure (JVP)

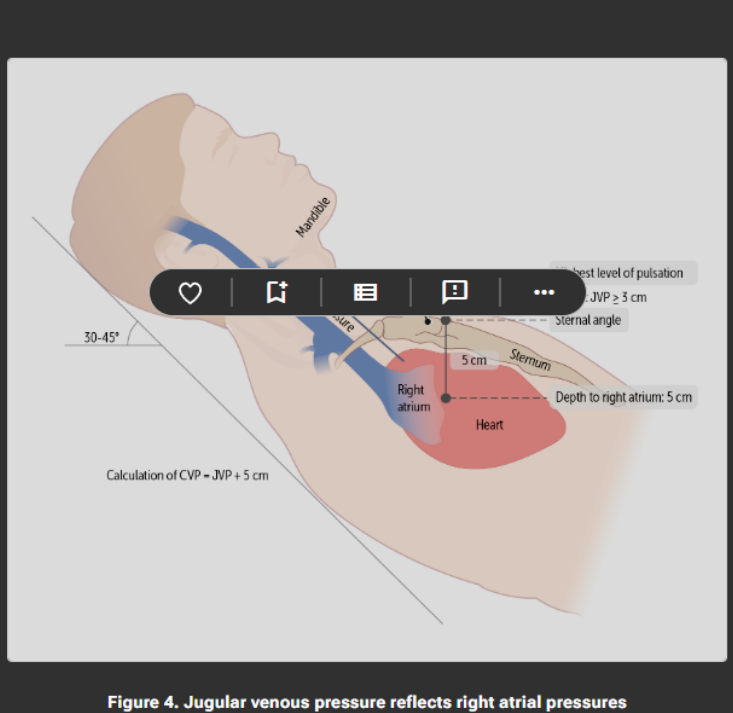

Mandible
30–45°
Highest level of pulsation
JVP ≥ 3 cm
Sternal angle
5 cm
Sternum
Right atrium
Depth to right atrium: 5 cm
Heart
Calculation of CVP = JVP + 5 cm

Figure 4. Jugular venous pressure reflects right atrial pressures

The **jugular venous pressure** provides a visible and palpable assessment of right atrial filling and pressure since the jugular vein drains directly into the right atrium Figure 4. Understanding jugular venous pressure (JVP), which can be visible at the bedside, is important in clinical practice because elevated JVP, called jugular venous distention (JVD) is seen in disorders that cause elevated **right atrial pressure (RAP)**.

Typically, you won't see actual dilated veins in JVD, since the internal jugular vein is deep in the neck. Instead, place the patient in the supine position with the torso at a 45-degree angle and look for a faint pulsation at the base of the neck. Measure the vertical distance from the sternal angle to the topmost point of the visible impulse to calculate the jugular venous pressure (Figure 4). The normal value is less than 3 cm.

A measured JVP ≥ 3 cm indicates elevated central venous pressure (CVP), often seen in right-sided he in inability to empty the right ventric oth right atrial and jugular venous pressure upstream.

Why is the jugular venous exam important?

Right Atrial Pressure (RAP) curves

Let's now look at the overall structure of the **right atrial pressure (RAP) curves** (Figure 5). The right atrial pressure curve also mirrors jugular venous pressure. The jugular venous pressure provides a visible and palpable reflection of right atrial pressure because the jugular vein drains directly into the right atrium.

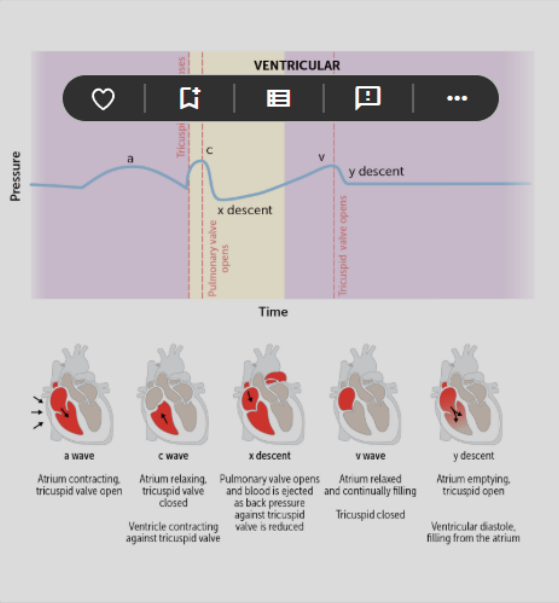

VENTRICULAR
Pressure
a
c
v
x descent
y descent
Time
a wave
c wave
x descent
v wave
y descent
Atrium contracting, tricuspid valve open
Atrium relaxing, tricuspid valve closed
Ventricle contracting against tricuspid valve
Pulmonary valve opens and blood is ejected as back pressure against tricuspid valve is reduced
Atrium relaxed and continually filling
Tricuspid closed
Atrium emptying, tricuspid open
Ventricular diastole, filling from the atrium

Figure 5. Atrial pressures during the cardiac cycle

a wave

The a wave represents a peak in pressure that results from **atrial contraction**. Atrial pressure is greatest in this part of the right atrial pressure curve and decreases when blood is expelled into the ventricle.

c wave

At the end of the a wave, the pressure drops when blood is expelled from the atrium. After this drop in pressure, the **ventricle contracts**, the pressure within it increases, and the tricuspid valve closes. Following this, there is a slight bump (called the c wave) that is caused by ventricular contraction against the closed tri essure. The contraction not only closes the valve but also pushes it back slightly into the right atrium. This pushing increases the pressure in the right atrium, and that produces the c wave. The c wave takes place during isovolumetric contraction of the cardiac cycle.

x descent

As the right ventricle continues to build up pressure, it eventually causes opening of the pulmonary valve. As this occurs, right ventricular pressure forces blood into the pulmonary system and reduces the back pressure into the right atrium. At the same time, **atrial relaxation** takes place. Both of these changes cause the drop in atrial pressure known as the x descent.

v wave

With the tricuspid valve now closed, venous return continues and the volume and pressure in the right atrium increase. The **atrial venous filling** causes a slow increase in pressure called the v wave. This occurs throughout ventricular contraction and into isovolumetric relaxation.

y descent

As the entire heart enters diastole (the period where both atria and ventricles are relaxed), the right-ventricular pressure falls (during isovolumetric relaxation) and the tricuspid valve opens. Now the blood that has been collecting in the right atrium while the tricuspid valve has been closed can passively enter the right ventricle. Right atrial volume and pressure thus decrease. This is referred to as the y descent and it occurs during the rapid **ventricular filling p**

Mnemonic

The letter names of the waves—**a**, **c**, **x**, and **v**—tell you when they happen: **a** wave = **A**trial contraction, **c** wave = ventri**C**ular **C**ontraction, **x** descent = atrial rela**X**ation, **v** wave = **V**enous filling, and **y** descent = empt**Y**ing of the atrium

CLINICAL CORRELATION

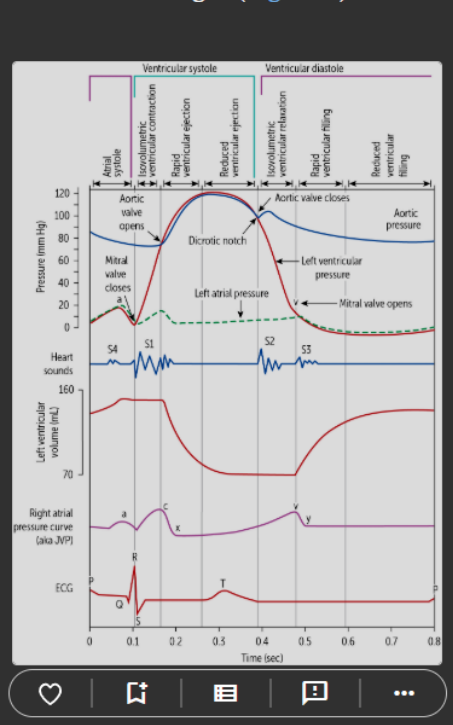
Aortic stenosis is a condition where the aortic valve becomes narrowed, making it harder for the left ventricle to eject blood. The stiffened valve delays opening, and results in a prolonged time the left ventricle is contracting without ejecting blood. This directly affects the ventricular ejection phase of the cardiac cycle, causing a prolonged **isovolumetric contraction phase** as the ventricle must generate higher pressure to overcome the stenotic valve resistance.

Relating this to the symptoms, less blood is ejected due to the obstruction, leading to low cardiac output symptoms (fatigue, syncope). The turbulent flow across the stenotic valve produces a heart sound/ murmur during ventricular ejection, with radiation to the carotids.

Putting It All Together

We've now covered many of the elements in the comprehensive Cardiac cycle or Wiggers diagram, named for the man who first summarized events of the cardiac cycle graphically on a time scale. Review this diagram thoroughly, mentally aligning each part and knowing why everything is happening.

This version adds in heart sounds and the ECG to the diagram of the cardiac cycle with pressure and volume changes (Figure 6).



One thing to notice about the ECG tracing is that the ECG impulse precedes the change in cardiac pressure—for example, the P wave starts before the increase in left atrial pressure, and the Q wave happens before the increase in LV pressure. This is because actual cardiac muscle contraction lags a bit behind the electrical impulse. So the ECG doesn't line up directly with the pressure and volume events.

CASE CONNECTION BACK TO INTRODUCTION ↑

Reflecting on Mr. J's condition, we now understand that his prolonged isovolumetric contraction phase occurs due to increased resistance from his narrowed aortic valve. His slow-rising pulse (pulsus parvus et tardus) and murmur during ventricular ejection correspond to this pathophysiology. By recognizing which phase of the cardiac cycle is disrupted, we can better interpret physical exam findings and determine appropriate interventions, such as echocardiography to assess valve function and possible valve replacement therapy.

Summary

- The cardiac valves prevent backflow during the cardiac cycle.
- The atria and ventricles are separated by the atrioventricular valves, and the ventricles and the great vessels are separated by the semilunar valves.
- During atrial systo rovide the final volume for ventricular fil
- During isovolumetric ventricular contraction, the initial pressure buildup closes the atrioventricular valves, and because the semilunar valves are also closed, there is an increase in ventricular pressure with no change in blood volume within the ventricles.
- During rapid ventricular ejection, the semilunar valves open and blood is rapidly ejected from the ventricles into the great vessels.
- During reduced ventricular ejection, the semilunar valves are still open but the rate of ejection is reduced.
- During isovolumetric ventricular relaxation, the initial pressure drop closes the semilunar valves, and because the atrioventricular valves are also closed, ventricular pressure drops with no change in blood volume within the ventricles.
- During rapid ventricular filling, the atrioventricular valves open and blood from the atria rapidly, but passively, fill the ventricles.
- During reduced ventricular filling, the atrioventricular valves are still open but the rate of passive filling is reduced.
- The pulse corresponds to rapid and reduced ventricular ejection.
- The first heart sound (S1) corresponds to the closure of the mitral and tricuspid valves.
- The second heart sound (S2) corresponds to the closure of the aortic and pulmonary valves.
- In a right atrial pressure curve, the a wave corresponds to atrial contraction, the c wave corresponds to ventricular contraction, the x descent results from ejection from the ventricle and relaxation of the atrium, the v wave results from venous filling against a closed atrioventricular valve, and the y descent is caused by the decrease in atrial volume when the

Review Questions

1. Aortic stenosis is a condition in which the aortic valve becomes calcified and stiff. The left ventricle has to generate more pressure to open the stiffer valve. When the valve opens, the increased ventricular pressure causes a jet of blood that radiates to the carotid arteries, which can be heard on auscultation. During which phase of the cardiac cycle are you most likely to hear the carotid radiation?

   A. ○ Atrial systole
   B. ○ Isovolumetric ventricular contraction
   C. ● Rapid ventricular ejection
   D. ○ Rapid ventricular filling
   E. ○ Reduced ventricular ejection

Hide Explanation

The correct answer is rapid ventricular ejection (C). In rapid ventricular ejection, the aortic valve opens, and blood is ejected from the ventricle and radiates to the carotid arteries in someone with aortic stenosis. No blood exits the ventricles during atrial systole (A). In isovolumetric ventricular contraction, the aortic valve has yet to open, so no blood will leave the left ventricle (B). The rapid ventricular filling phase (D) encompasses ventric e heard; neither would this occur during reduced ventricular ejection (E).

2. During which phase of the cardiac cycle are the semilunar valves open but the atrioventricular valves are closed?

   A. ○ Atrial systole
   B. ○ Isovolumetric ventricular contraction
   C. ○ Isovolumetric ventricular relaxation
   D. ● Rapid ventricular ejection
   E. ○ Rapid ventricular filling

Hide Explanation

The correct answer is rapid ventricular ejection (D). During this phase, the semilunar valves are open, permitting blood to flow forward from the ventricles to the great vessels. The high ventricular pressure during this phase keeps the atrioventricular valves closed, preventing backflow into the atria. The reverse is true during atrial systole (A) and rapid ventricular filling (E), when the semilunar valves are closed and the atrioventricular valves are open to allow the filling of the ventricles. During the isovolumetric phases (B, C), both valve types are closed.

3. Atrial fibrillation is ntract effectively. If a right atrial pressure curve is obtained from a patient with atrial fibrillation, which of the following would be absent?

   A. ● a wave
   B. ○ c wave
   C. ○ v wave
   D. ○ x descent
   E. ○ y descent

Hide Explanation

The correct answer is a wave (A). The "a" wave results from atrial contraction, and during atrial fibrillation, the atria's inability to effectively contract means that it cannot elevate the right atrial pressure enough to produce a visible wave. The x descent is present but attenuated (D); since atrial contraction does not occur, atrial relaxation cannot occur either. The other components of the pressure curve are unaffected: c wave (B), v wave (C), and y descent (E).

4. Which of the following effects on the right atrial pressure curve is produced by venous filling?

   A. ○ a wave
   B. ○ c wave
   C. ● v wave
   D. ○ x descent
   E. ○ y descent

Hide Explanation

The correct answer is v wave (C). The v wave results from the pressure of venous blood filling into the right atrium. The a wave results from atrial contraction (A). The c wave results from ventricular contraction (B). The x descent results from ventricular emptying and atrial relaxation (D). The y descent results from the tricuspid valve opening and blood entering the ventricle (E).

How would you rate this Brick?

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Explain systole, diastole, and relate to the function of the atrioventricular valves, semilunar valves, and great vessels.~~
02 ☑ ~~Explain how the pressure and volume in the heart chambers and major vessels change in each of the seven phases of the cardiac cycle.~~
03 ☑ ~~Correlate the cardiac cycle phases with these physical exam findings: pulse, first and second heart sounds, and the wave forms of the atrial pressure.~~

Objective Confidence: 3/3